Food and Agriculture
Organization of the
United Nations

Technical workshop for GCF project: Climate change and climate change impacts on São Tomé and Príncipe with a focus on fishery and tourism sectors in a Blue Economy context

Sao Tome and Principe, November/December 2025

IMPACT MODELLING FOR FISH BIOMASS PROJECTIONS

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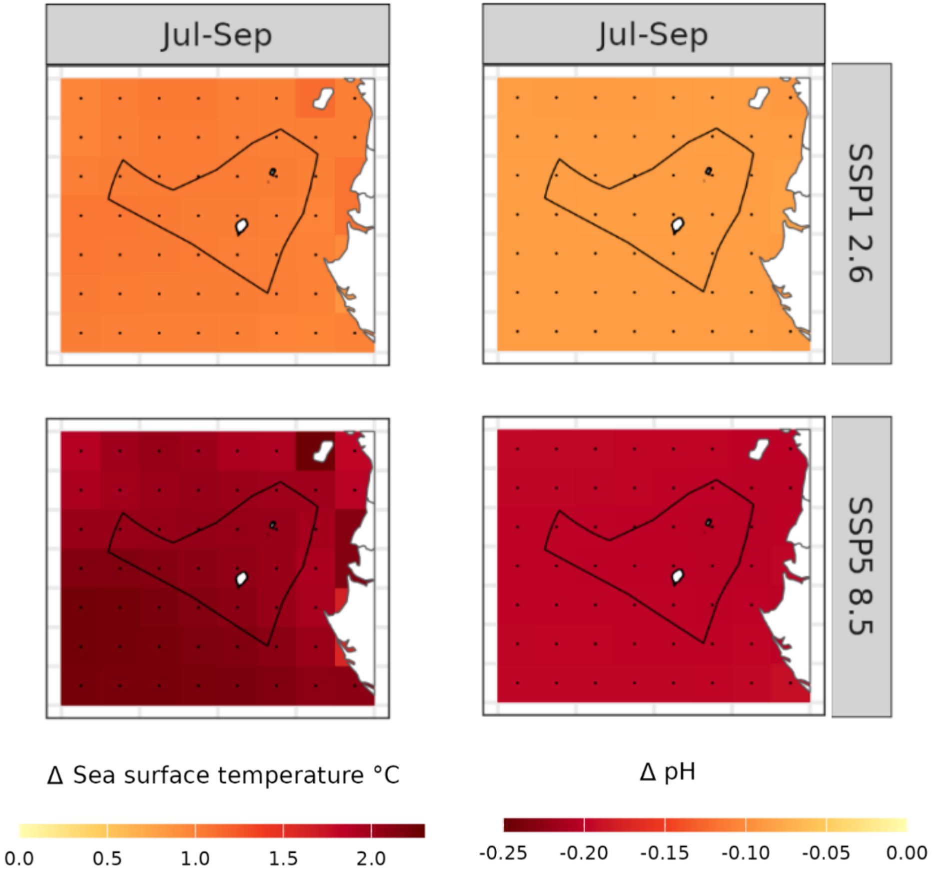
Fisheries in Sao Tome and Principe: the challenge

- Climate models project significant future oceanographic change
- Climate change is expected to affect fisheries by altering:
  - Fish reproduction
  - Fish densities
  - Fish migration patterns



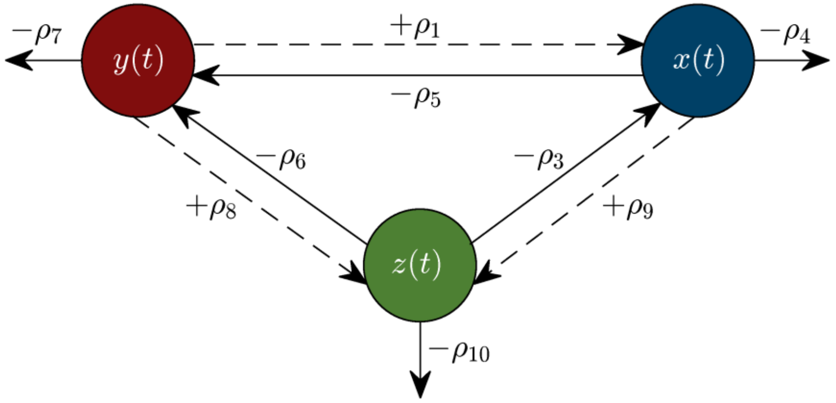
Values are relative to historical run

Impact-based modelling

- **Impact models** are referred to as "mechanistic" as they use complex equations to model how fish systems function and interact with their environment

- For example, impact models often consider:
    - Predator-prey interactions
    - Resource availability
    - Energy transfer
    - Migration

- Impact models give projections of fish biomass based on future projected climate inputs



The Inter-Sectoral Impact Model Intercomparison Project

- ISIMIP is a project aimed at providing a consistent framework for climate impact modelling
- Specific focus topics:
  - Agriculture
  - Water
  - Forests
  - Fisheries
  - Energy
- Input:
  - Specific climate model ensemble (CMIP) and Earth System Model projections
  - Specific climate forcings (SSPs)
  - Consistent impact models
- Output:
  - Global impact model simulations
  - Climate/ocean forcing inputs can also be extracted

ISIMIP
Inter-Sectoral Impact Model Intercomparison Project

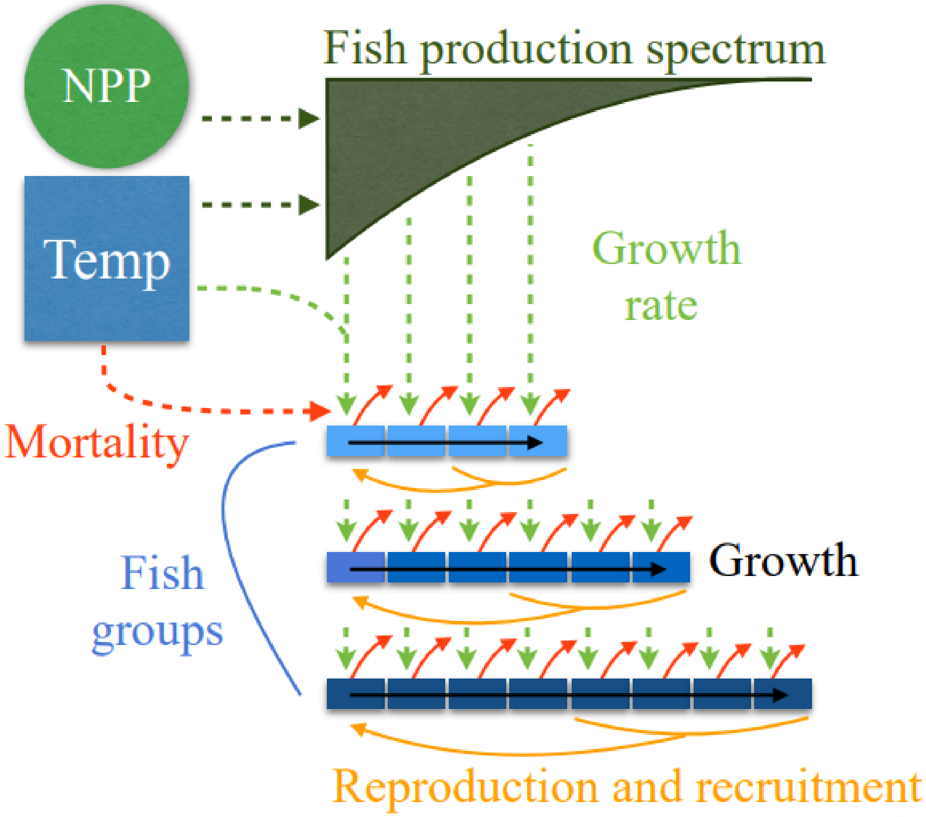
Fish biomass impact models

Available fish biomass impact models:

- APECOSM
- BOATS (fishing intensity)
- EcoOcean (fishing intensity)
- DBPM
- FEISTY
- ZooMSS

- Climate models
- IPSL-CMIP6-LR
- GFDL-ESM4

BiOeconomic mArine Trophic Size-spectrum (BOATS)



Simulating climate impact on fish biomass

Available fish biomass impact models:

- APECOSM
- BOATS (fishing intensity)
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- FEISTY
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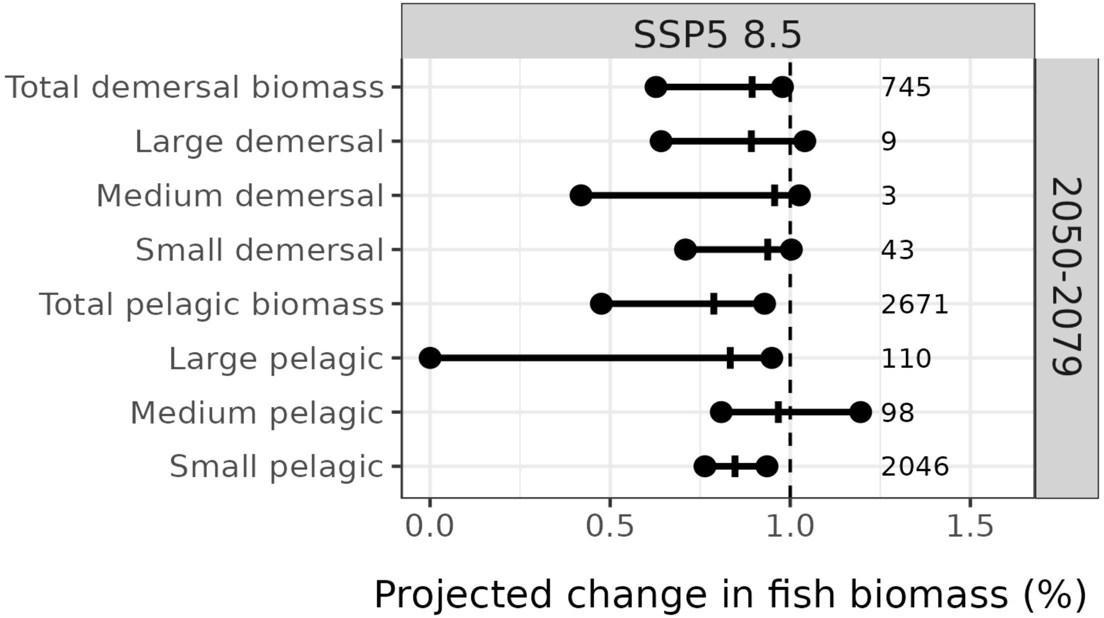

Incorporation of fishing intensity into models

Available fish biomass impact models:

- APECOSM
- BOATS (**fishing intensity – total only**)
- EcoOcean (**fishing intensity**)
- DBPM
- FEISTY
- ZooMSS

- Climate models
- IPSL-CMIP6-LR
- GFDL-ESM4